Structure Learning of Sparse GGMs Over Multiple Access Networks

Mostafa Tavassolipour, Armin Karamzade, Reza Mirzaeifard, Seyed Abolfazl Motahari, and Mohammad-Taghi Manzuri Shalmani

***Abstract*— A central machine is interested in estimating the underlying structure of a sparse Gaussian Graphical Model (GGM) from a dataset distributed across multiple local machines. The local machines can communicate with the central machine through a wireless multiple access channel. In this paper, we are interested in designing effective strategies where reliable learning is feasible under power and bandwidth limitations. Two approaches are proposed: Signs and Uncoded methods. In the Signs method, the local machines quantize their data into binary vectors and an optimal channel coding scheme is used to reliably send the vectors to the central machine where the structure is learned from the received data. In the Uncoded method, data symbols are scaled and transmitted through the channel. The central machine uses the received noisy symbols to recover the structure. Theoretical results show that both methods can recover the structure with high probability for a large enough sample size. Experimental results indicate the superiority of the Signs method over the Uncoded method under several circumstances.**

***Index Terms*— Structure learning, Gaussian graphical model, distributed learning.**

I. Introduction

In A Probabilistic Graphic Model (PGM), the dependency structure of random variables is encoded by a graph. Structure learning refers to the recovery of such graphs based on samples from the model. Structure learning of PGMs has been used in many fields including biology, information technology, and machine learning. Being one of the most applicable instances of PGMs, the Gaussian Graphical Model (GGM) represents the dependency structure of jointly Gaussian random variables. Inferring the structure of GGMs from samples has applications such as modeling gene regulatory networks [1], [2], analyzing social networks [3], modeling brain connectivities [4], [5], speech recognition [6].

Several methods and algorithms are proposed for structure learning of GGMs from samples, c.f. [7]–[12]. When the underlying structure is sparse, methods based on *lasso*[1] [13]–[15] have attracted many researchers. In the lasso-based structure learning, an ℓ_1-regularized maximum likelihood of the precision matrix is solved [16]–[19]. The lasso-based methods are analyzed in several articles (see [20] and refs therein). For instance, Ravikumar *et al.* [20] analyzed the performance of the ℓ_1-regularized maximum likelihood estimator (MLE) in high dimensional scaling. They showed that with probability converging to one, the estimated structure correctly specifies the zero pattern of the true precision matrix.

Besides the lasso-based estimators, several thresholding-based estimators are proposed for sparse recovering of the precision matrix [21], [22]. For example, Sojoudi [21] proposed a simple thresholding method and showed, under certain conditions, the resulting structure is identical to the structure obtained by the lasso. In a particular case where the underlying graph structure is a tree, Chow-Liu algorithm [23] can recover the maximum likelihood structure. Tan *et al.* in [24] and [25] provided an analysis of the error exponent of the Chow-Liu algorithm.

In many cases, samples are distributed over several hosting machines. In such settings, the conventional method is to gather the samples in a central machine to perform the inference task. However, transferring the local datasets is not possible in situations where datasets are massive or communication is very limited. There are many works addressing various statistical inference problems in distributed settings [26]–[29]. However, the problem of the structure estimation of GGM over distributed data is less studied in the literature.

Distributed estimation of the underlying structure has various applications such as security, surveillance, and personal health care monitoring [30]. In such scenarios, nodes sense the environment and transmit their data to a central node where data are analyzed and the structure is inferred. Since spatially distributed sensor nodes have limited communication power, designing a communication-efficient method for learning of the structure in distributed settings is of great importance. Distributed structure learning previously studied in several papers from various perspectives [31]–[35]. For example, Tavassolipour *et al.* [35] proposed a method based on the Chow-Liu algorithm to estimate the structure in distributed settings. Meng *et al.* [32] addressed the estimation of the precision matrix in a distributed manner where the zero-pattern of the precision matrix is known in advanced.

In this paper, we address the problem of structure learning of GGMs in distributed environments. We assume that the data are distributed over multiple local machines so that each one contains a single dimension of the whole dataset.

Manuscript received May 16, 2019; revised September 21, 2019; accepted November 13, 2019. Date of publication December 2, 2019; date of current version February 14, 2020. The associate editor coordinating the review of this article and approving it for publication was B. Shim. *(Corresponding author: Seyed Abolfazl Motahari.)*

The authors are with the Computer Engineering Department, Sharif University of Technology, Tehran 11365, Iran (e-mail: tavassolipour@ce.sharif.edu; karamzade@ce.sharif.edu; mirzaeifard@ce.sharif.edu; motahari@sharif.edu; manzuri@sharif.edu).

Color versions of one or more of the figures in this article are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TCOMM.2019.2957080

[1]Least Absolute Shrinkage and Selection Operator.

0090-6778 © 2019 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on February 07,2026 at 01:36:22 UTC from IEEE Xplore. Restrictions apply.

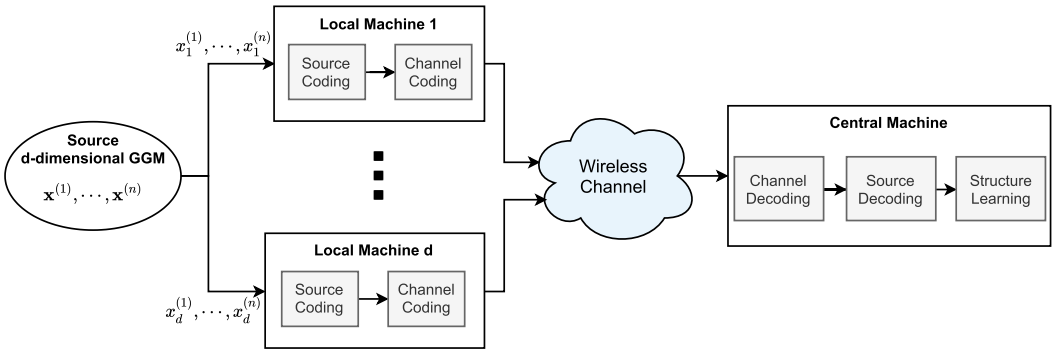


Fig. 1. The whole system block diagram.

The local machines are connected to a central machine via a wireless medium. The central machine is responsible for inferring the conditional dependencies between the gathered data by the local machines. The system's block diagram is depicted in Fig. 1. We also assume that the communication links between the local and the central machine are bandwidth limited implying that transmission of the whole local datasets to the central machine is impossible. Due to this constraint, each local machine transmits some information from its local dataset to the central machine. Then, the central machine estimates the underlying graph structure using received information from the local machines. Arroyo and Hou [34] studied the problem of sparse precision matrix estimation in a situation where the samples are distributed among several machines. Their work differs from our setting in the sense that we assume the data are split across dimensions, whereas they split the data across samples.

We have proposed two communication schemes for transmitting information from the local machines to the central machine. In the first scheme, we have separated the source coding from the channel coding. In this scheme, we quantize the source samples into single bits and assume that there exists a channel coding such that the bits can be sent through the channel reliably. We refer to this scheme as *Signs method*.

In the second scheme, we do not use any source and channel coding. In this scheme, we put the source samples into the channel without any encoding. At the central machine, we directly estimate the underlying graph structure using received symbols from the channel. We refer to this scheme as *Uncoded method*.

We have shown through theoretical analysis and experiments that by transmitting only 1 bit per sample, the central machine can reliably recover the underlying graph structure. More precisely, we have shown theoretically that by consuming only 1 bit per sample, under some mild conditions, the central machine can perfectly recover the graph structure with high probability. Moreover, the true signs of the edges' weights of the graph are obtained.

The paper is organized as follows. Section II describes the detail of our modeling for the source and the communication channel. In sections III and IV, we describe Signs and Uncoded methods, respectively. Section V provides experimental results to compare and evaluate the proposed methods. Finally, Section VI concludes the paper.

II. Problem Formulation

A GGM for a d-dimensional random vector $\mathbf{x} \in \mathbb{R}^d$ is specified by a graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$ where $\mathcal{V} = \{1, \cdots, d\}$ is the set of vertices and $\mathcal{E} \subseteq \mathcal{V}^2$ is the set of edges. The model comprises all d-dimensional normal distributions $\mathcal{N}(\boldsymbol{\mu}, Q_x)$ where Q_x is the covariance matrix and $\Theta_x = Q_x^{-1}$ is the precision matrix with entries $(\Theta_x)_{jk} \neq 0$, iff $(j,k) \in \mathcal{E}$.

A central machine is interested in reconstructing $\mathcal{G}(\mathcal{V}, \mathcal{E})$, from n i.i.d. random samples $\{\mathbf{x}^{(1)}, \cdots, \mathbf{x}^{(n)}\}$ which are drawn from the model and distributed across d separate local machines. Each local machine has access to a single dimension of all samples. More specifically, the data available at the j-th local machine is $\{x_j^{(1)}, \cdots, x_j^{(n)}\}$ where $x_j^{(i)}$ denotes the j-th dimension of the i-th sample. In this situation, the central machine needs to communicate with the local machines to acquire enough information for reconstructing the graph. For convenience, we assume that the data at each local machine has zero mean and unit variance, i.e. $\mu_j = 0$ and $(Q_x)_{jj} = 1$.

To transmit information, each local machine is equipped with a single antenna and constrained to a transmit power p. The central machine receives information by incorporating d receive antennas. The channel between the local machines and the central node is assumed to be a wireless multiple access channel which can be modeled as

$$\mathbf{y}(t) = H\mathbf{s}(t) + \mathbf{z}(t), \tag{1}$$

where $\mathbf{y}(t)$ and $\mathbf{s}(t) = [s_1(t), \cdots, s_d(t)]^T$ are the received and transmit vectors at time t, respectively. $H \in \mathcal{C}^{d \times d}$ is a complex matrix that contains the channel gains which are drawn from independent circularly symmetric complex Gaussian distribution. $\mathbf{z}(t)$ is an independent Gaussian vector with covariance matrix $\sigma_z^2\ I_d$, where I_d is the $d \times d$ identity matrix. Throughout the paper, we assume that the channel parameters (H and σ_z^2) are known at the receiver and H is invertible.

Two separate communication strategies, namely coded and uncoded, are considered to send information to the central machine. In the coded scheme, samples available at each transmitter are encoded by information bits and then a channel coding scheme is incorporated to convey these bits simultaneously and reliably to the central node. In the uncoded scheme, on the other hand, samples are converted directly to input symbols at each transmitter and the central machine uses the

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on February 07,2026 at 01:36:22 UTC from IEEE Xplore. Restrictions apply.

received symbols directly to reconstruct the graph. In either case, the central node attempts to estimate the covariance matrix from the received information.

Assuming availability of the samples at the central node, the sample covariance matrix can be computed as $S_x = \frac{1}{n}\sum_{i=1}^{n} \mathbf{x}^{(i)} \left(\mathbf{x}^{(i)}\right)^T$ and the negative log-likelihood is given by

$$g(\Theta) = \mathrm{tr}(\Theta S_x) - \log\det(\Theta). \tag{2}$$

The maximum likelihood estimate of the precision matrix is simply S_x^{-1}. Having some constraints on the precision matrix, the inverse sample covariance matrix is not efficient. In sparse precision matrix cases, the following program which is the ℓ_1-regularized version of (2) is proposed:

$$\widehat{\Theta} = \arg\min_{\Theta \succ 0} \left\{ g(\Theta) + \lambda_n \|\Theta\|_{1,\mathrm{off}} \right\}, \tag{3}$$

where $\|\Theta\|_{1,\mathrm{off}} := \sum_{j\neq k} |\Theta_{jk}|$, and λ_n is a user defined regularization parameter. There is an efficient algorithm known as *graphical lasso (glasso)* [16] for solving the above program.

The central machine, however, does not have access to S_x and needs to obtain another estimate of the covariance matrix based on the received symbols. This is the main objective of this paper in either coded and uncoded schemes. In fact, we propose two strategies, namely Sings and Uncoded methods, which specify the ways of communicating information together with estimating the covariance matrix at the central machine in both cases.

A. Incoherence Condition

It is worth mentioning that, in general, solving the optimization problem in (3) does not necessarily recover the underlying graph structure even if the sample size approaches to the infinity, see [36] for some impossible cases. In fact, [36] shows that changing the regularization parameter cannot resolve the problem either. Hence, some constraints on $g(\Theta)$ are needed to attain the consistency of the solution. Here, we adopt the incoherence condition used by Ravikumar *et al.* [20] to obtain error bounds on the consistency of the solutions.

Before stating the incoherence condition, we need several definitions and notations. In [20], it is stated that the Hessian matrix of (2) is given by

$$\Gamma = \nabla^2_{\Theta} g(\Theta)\Big|_{\Theta=\Theta_x} = \Theta_x^{-1} \otimes \Theta_x^{-1}, \tag{4}$$

where $\otimes$ is the Kronecker matrix product. The entry $\Gamma_{(j,k),(l,m)}$ corresponds to the second derivative $\frac{\partial^2\, g}{\partial\Theta_{jk}\partial\Theta_{lm}}$, evaluated at $\Theta = \Theta_x$ and $\Gamma_{(j,k),(l,m)} = \mathrm{cov}\{x_j x_k, x_l x_m\}$.

We define the set of non-zero entries in the true precision matrix Θ_x as

$$E(\Theta_x) = \{(j,k) \in \mathcal{V}^2 | (\Theta_x)_{jk} \neq 0\}, \tag{5}$$

where $\mathcal{V} = \{1, \cdots, d\}$. Note that $E(\Theta_x)$ includes the diagonal entries. Let $E^c(\Theta_x)$ be the complement set of $E(\Theta_x)$ which includes all pairs (j,k) where $(\Theta_x)_{jk} = 0$. For any two subsets T and T' of $\mathcal{V}\times\mathcal{V}$, the notation $\Gamma_{TT'}$ denotes a $|T|\times|T'|$ sub-matrix of Γ with rows and columns indexed by T and T', respectively.

The max-row-sum norm of a d by d matrix X is denoted by $|||X|||_\infty$ and defined as

$$|||X|||_\infty = \max_{i=1,\cdots,d} \sum_{j=1}^{d} |X_{ij}|. \tag{6}$$

We define two quantities κ_Σ and κ_Γ as

$$\kappa_\Sigma := |||Q_x|||_\infty, \tag{7}$$

$$\kappa_\Gamma := |||\Gamma_{EE}^{-1}|||_\infty. \tag{8}$$

These quantities are appeared in the error bounds of the proposed methods.

Aassumption 1 (Incoherence Condition): There exists some $\alpha \in (0,1]$ *such that*

$$|||\Gamma_{E^cE}(\Gamma_{EE})^{-1}|||_\infty \leq (1-\alpha). \tag{9}$$

An implication of the above condition is that the non-edge pairs cannot have a strong influence on the edges.

III. Signs Method

In this section, we propose a simple method in estimating the graph structure of the GGMs in our distributed data setting. This method consists of four steps. First, local machines apply the sign function to their local datasets. Second, all the signs are transmitted to the central machine using appropriate channel coding schemes. Third, an estimate of the sample covariance matrix S_x is obtained from the received signs. Finally, the optimization problem (3) is solved to obtain the structure of the underlying graph. To make sure the second step is feasible, the channel capacity between the local machines and the central machine must be greater than or equal to 1.

The focus of this section is the third step where an estimator for the sample covariance matrix from the signs is proposed. We also provide theoretical guarantees for the proposed estimator.

Let x_j and x_k be jointly normal with zero means, unit variances and correlation coefficient ρ_{jk}. If $\hat{x}_j$ and $\hat{x}_k$ be the corresponding sign variables, then the joint probability mass function (pmf) of $\hat{x}_j$ and $\hat{x}_k$ is given by [37]

$\hat{x}_j \backslash \hat{x}_k$	-1	$+1$
-1	$\beta_{jk}/2$	$(1-\beta_{jk})/2$
$+1$	$(1-\beta_{jk})/2$	$\beta_{jk}/2$

(10)

where $\beta_{jk} \in [0,1]$ and is given by

$$\beta_{jk} = \frac{1}{2} + \frac{\arcsin(\rho_{jk})}{\pi}. \tag{11}$$

The equation (11) can be rewritten as

$$\rho_{jk} = \sin(\pi(\beta_{jk} - \frac{1}{2})) = -\cos(\pi\beta_{jk}). \tag{12}$$

Thus, by proposing an estimator for β_{jk}, using (12) we can obtain an estimator for ρ_{jk}. The following estimator for β_{jk} is optimal in the sense that it is unbiased and has minimum variance (UMVE) [29],

$$\hat{\beta}_{jk} = \frac{1}{n}\sum_{i=1}^{n} \mathbb{I}(\hat{x}_j^{(i)}\hat{x}_k^{(i)} = 1), \tag{13}$$

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on February 07,2026 at 01:36:22 UTC from IEEE Xplore. Restrictions apply.

where $\mathbb{I}(.)$ is the indicator function. By substituting $\hat{\beta}_{jk}$ in (12), we use

$$\hat{\rho}_{jk} = -\cos(\pi\hat{\beta}_{jk}), \tag{14}$$

as an estimator for ρ_{jk}. Although $\hat{\rho}_{jk}$ is biased, it is a consistent estimator for ρ_{jk}. Following lemma gives an upper bound on the error probability of this estimator.

Lemma 1: Let x_j and x_k be two jointly normal variables with zero means, unit variances and correlation coefficient ρ_{jk}. Then, for the estimator $\hat{\rho}_{jk}$ in (14)*, we have*

$$\Pr\left(|\hat{\rho}_{jk} - \rho_{jk}| \geq \delta\right) \leq 2\exp\left(-\frac{2}{\pi^2}n\delta^2\right), \tag{15}$$

where $\delta \geq 0$.

Proof: Since the function $\cos(\cdot)$ is a 1-Lipschitz function, i.e. $|\cos(x) - \cos(y)| \leq |x - y|$, we have

$$\begin{aligned}\Pr\left(|\hat{\rho}_{jk} - \rho_{jk}| \geq \delta\right) &= \Pr\left(|-\cos(\pi\hat{\beta}_{jk}) + \cos(\pi\beta_{jk})| \geq \delta\right)\\ &\leq \Pr\left(\pi|\hat{\beta}_{jk} - \beta_{jk}| \geq \delta\right)\\ &= \Pr\left(|\hat{\beta}_{jk} - \beta_{jk}| \geq \frac{\delta}{\pi}\right).\end{aligned}$$

Applying the Hoeffding inequality yields

$$\Pr\left(|\hat{\beta}_{jk} - \beta_{jk}| \geq \frac{\delta}{\pi}\right) \leq 2\exp\left(-\tfrac{2}{\pi^2}n\delta^2\right),$$

which completes the proof. □

Lemma 1 shows an exponential decay in the error probability of $\hat{\rho}_{jk}$ with respect to the sample size. Using this estimator, we can obtain an estimator for the sample covariance matrix as follows

$$\widehat{S}_x = -\cos(\pi B), \tag{16}$$

where $B_{jk} = \hat{\beta}_{jk}$, and $\cos(.)$ function is applied on the input matrix element-wise, i.e. $(\widehat{S}_x)_{jk} = -\cos(\pi B_{jk})$. Note that $\widehat{S}_x$ has entries equal to 1 at its diagonal, since $\hat{\beta}_{jj} = 1$. This means that there is no error in the variance estimation as it was known a priori.

Note that the sample covariance matrix $\widehat{S}_x$ defined in (16) is not necessarily positive semi-definite. However, it does not affect the convexity and uniqueness of the solution of (3). This comes from the result presented in [20] which shows that (3) is convex and has a unique solution for any sample covariance matrix with strictly positive diagonal elements which holds for $\widehat{S}_x$ in (16).

Let Θ_x and $\widehat{\Theta}_x$ be the true and estimated precision matrices, respectively. Then, $\mathcal{M}(\Theta_x; \widehat{\Theta}_x)$ is defined as the event indicating Θ_x and $\widehat{\Theta}_x$ are matched in their zero entries and have the same sign in their nonzero entries. Theorem 1 shows that the event $\mathcal{M}(\Theta_x, \widehat{\Theta}_x)$ occurs with high probability for the Signs method described above. Before stating the theorem we should define some properties of the underlying GGM. We denote the maximum degree of the underlying graph structure by Δ. The minimum absolute value of the nonzero elements in the precision matrix is denoted by $\theta_{\min}$ which is

$$\theta_{\min} := \min_{(i,j)\in E(\Theta_x)} |\,(\Theta_x)_{ij}\,|. \tag{17}$$

The following lemma from [20] shows that if there exists a sample covariance matrix S_x which satisfies some tail conditions, then the solution of the optimization program (3) recovers the zero-pattern of the precision matrix with high probability.

Lemma 2 (Ravikumar et al. [20]): Consider a normal distribution satisfying Assumption 1 with parameter $\alpha \in (0, 1]$. Let S be a sample covariance matrix such that for any $(j, k) \in \mathcal{V}^2$:

$$\Pr\left(|S_{jk} - Q_{jk}| \geq \delta\right) \leq f(n, \delta) \quad \textit{for all } \delta \in (0, 1/\nu_*], \tag{18}$$

where Q is the true covariance matrix, f is a monotonically decreasing function in n and δ, and ν_* is a positive constant indicating the maximum possible value of δ. We define two inverse functions as

$$\bar{n}(\delta, \epsilon) = \max\left\{n : \Pr\left(|S_{jk} - Q_{jk}| \geq \delta\right) \geq \epsilon\right\}, \tag{19}$$

$$\bar{\delta}(n, \epsilon) = \max\left\{\delta : \Pr\left(|S_{jk} - Q_{jk}| \geq \delta\right) \geq \epsilon\right\}. \tag{20}$$

Let $\widehat{\Theta}$ be the solution of the optimization problem (3) *with $S_x = S$ and regularization parameter $\lambda_n = \frac{8}{\alpha}\bar{\delta}(n, \epsilon)$. Then with probability at least $1 - d^2\epsilon$ for any $0 < \epsilon \leq d^{-2}$,*

(a) The edge set specified by $\widehat{\Theta}$ is a subset of the true edge set if

$$n > \bar{n}\left(\frac{1}{\max\{\nu_*, 6(1+8\alpha^{-1})\Delta\max\{\kappa_\Sigma\kappa_\Gamma, \kappa_\Sigma^3\kappa_\Gamma^2\}\}}, \epsilon\right). \tag{21}$$

(b) The edge set specified by $\widehat{\Theta}$ is consistent with the true edge set if $n > \bar{n}()$, (22), shown at the bottom of this page.

Proof: For the proof, see theorems 1 and 2 in [20]. □

Lemma 1 and 2 are essentials in obtaining the following theorem which shows that one can use the estimator (16) and the log-determinant program (3) to recover the underlying graph structure with high probability.

Theorem 1: Consider a normal distribution satisfying Assumption 1 with parameter $\alpha \in (0, 1]$. Let $\widehat{\Theta}_x$ be the solution of the log-determinant program (3) *with the sample covariance $\widehat{S}_x$ defined in* (16) *and regularization parameter $\lambda_n = (8\pi/\alpha)\sqrt{\frac{1}{2n}\ln\frac{2}{\epsilon}}$ for any $0 < \epsilon \leq d^{-2}$.*

(a) If the sample size is lower bounded as

$$n > C_{\text{sign}}^2 \ \ln\frac{2}{\epsilon}, \tag{23}$$

where

$$C_{\text{sign}} = 3\sqrt{2}\pi\ \Delta\left(1 + \frac{8}{\alpha}\right)\kappa_\Sigma^3\kappa_\Gamma^2,$$

$$n > \bar{n}\left(\frac{1}{\max\{2(1+\frac{8}{\alpha})\kappa_\Gamma\theta_{\min}^{-1}, \nu_*, 6(1+\frac{8}{\alpha})\Delta\max\{\kappa_\Sigma\kappa_\Gamma, \kappa_\Sigma^3\kappa_\Gamma^2\}\}}, \epsilon\right). \tag{22}$$

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on February 07,2026 at 01:36:22 UTC from IEEE Xplore. Restrictions apply.

then with probability at least $1-d^2\epsilon$, the edge set specified by $\widehat{\Theta}_x$ is a subset of the true edge set.

(b) If the sample size satisfies the lower bound

$$n > T_{\text{sign}}^2 \ln \frac{2}{\epsilon}, \tag{24}$$

where

$$T_{\text{sign}} = \max\left\{\sqrt{2}\pi\kappa_\Gamma\theta_{\min}^{-1}\left(1+\frac{8}{\alpha}\right), C_{\text{sign}}\right\},$$

then,

$$\Pr\left(\mathcal{M}(\widehat{\Theta}_x;\Theta_x)\right) \geq 1-d^2\epsilon. \tag{25}$$

Proof: We only need to combine Lemma 1 and 2 to obtain the desired result. In fact, $f(n,\delta)$, the key function in Lemma 2, can be obtained from Lemma 1 by observing that

$$\begin{aligned}\Pr\left(|(\widehat{S}_x)_{jk}-(Q_x)_{jk}| \geq \delta\right) &= \Pr\left(|\hat{\rho}_{jk}-\rho_{jk}| \geq \delta\right)\\ &\leq 2\exp\left(-\frac{2}{\pi^2}n\delta^2\right)\\ &\triangleq f(n,\delta),\end{aligned}$$

where $\delta \in (0,+\infty]$. Hence, $\nu_* = 0$ and the inverse functions $\bar{\delta}(n,\epsilon)$ and $\bar{n}(\delta,\epsilon)$ can be derived as

$$\bar{\delta}(n,\epsilon) = \pi\sqrt{\frac{1}{2n}\ln\frac{2}{\epsilon}}, \tag{26}$$

$$\bar{n}(\delta,\epsilon) = \frac{\pi^2}{2\delta^2}\ln\frac{2}{\epsilon}. \tag{27}$$

Substituting the above inverse functions into part *(a)* of Lemma 2 yields the sample size lower bound as

$$n > \frac{\pi^2}{2}\left(\max\left\{\nu^*, 6\left(1+8\alpha^{-1}\right)\Delta\max\left\{\kappa_\Sigma\kappa_\Gamma, \kappa_\Sigma^3\kappa_\Gamma^2\right\}\right\}\right)^2\ln\frac{2}{\epsilon}. \tag{28}$$

Since $(Q_x)_{ii} = 1$ for $i \in \{1,\cdots,d\}$, we have

$$\kappa_\Sigma = |||Q_x|||_\infty \geq 1. \tag{29}$$

Moreover, since $\text{trace}(Q_x) = d$, sum of eigenvalues of Q_x is d. Due to the arithmetic mean-harmonic mean inequality, the sum of inverse of the eigenvalues (or equivalently, the trace of the precision matrix) is at least d. This implies that at least one of the elements on the diagonal of Θ_x is greater than or equal to one. Thus, we have

$$\begin{aligned}\kappa_\Gamma &= |||\Gamma_{EE}^{-1}|||_\infty = |||\Theta_x\otimes\Theta_x|||_\infty\\ &= |||\Theta_x|||_\infty^2 \geq \max_i(\Theta_x)_{ii} \geq 1,\end{aligned} \tag{30}$$

Substituting the inequalities (29) and (30) into (28) yields

$$n > \left(3\sqrt{2}\pi\left(1+8\alpha^{-1}\right)\Delta\kappa_\Sigma^3\kappa_\Gamma^2\right)^2\ln\frac{2}{\epsilon},$$

which completes the proof of part *(a)* of the theorem. Similarly, the part *(b)* is resulted by substituting the inverse functions (26) and (27) into part *(b)* of Lemma 2. □

Theorem 1 shows that if the sample complexity scales with $\ln(1/\epsilon)$, then the Signs method can recover the graph structure with probability at least $1-d^2\epsilon$. The constants C_{sign} and T_{sign} depends on the underlying GGM parameters (i.e. $\Delta, \kappa_\Sigma, \kappa_\Gamma$) and the parameter α defined in Assumption 1. According to part (a) of the theorem, since $C_{\text{sign}} \leq T_{\text{sign}}$, the method has no false positive (predicting non-edges as edge) with small sample sizes. In Section V, we experimentally show that the false positive rate of the proposed Signs method decays rapidly with sample size.

Remark 1: Note that the proposed Signs method is applicable for any channel with capacity greater than or equal to 1 bit.

IV. Uncoded Method

In this section, we assume that each local machine puts its local data into the channel without any source or channel coding. The central machine estimates the underlying graph structure using the received data from the channel. At the central machine, no source or channel decoding is used. It infers the structure directly from the output of the channel.

According to the channel model described in Section II, each local machine can transmit two consequent samples by each channel use. More precisely, denoting the channel input symbol by $\mathbf{s} = \mathbf{s}_R + j\mathbf{s}_I$, each local machine can put two consequent samples as real and imaginary parts of the input symbol. We assumed that the number of receivers at the central machine is the same as the dimension of the normal distribution (i.e. d). Therefore, at the central machine, $n/2$ vectors are received that each one is $2d$-dimensional.

The equation (1) can be decomposed as

$$\mathbf{y} = (H_R + jH_I)(\mathbf{s}_R + j\mathbf{s}_I) + (\mathbf{z}_R + j\mathbf{z}_I), \tag{31}$$

where $j^2 = -1$. Hence, it can be rewritten in a block-matrix form as

$$\begin{bmatrix}\mathbf{y}_R\\ \mathbf{y}_I\end{bmatrix} = \begin{bmatrix}H_R & -H_I\\ H_I & H_R\end{bmatrix}\begin{bmatrix}\mathbf{s}_R\\ \mathbf{s}_I\end{bmatrix} + \begin{bmatrix}\mathbf{z}_R\\ \mathbf{z}_I\end{bmatrix}, \tag{32}$$

where H_R and H_I are $d\times d$ real matrices, and all the real and imaginary part vectors are d-dimensional.

Using both real and imaginary parts, two source samples can be transmitted per each channel use: a sample is put into the real part and the other is put into the imaginary part. In particular, $\mathbf{s}_R + j\mathbf{s}_I = \sqrt{p/2}(\mathbf{x}_R + j\mathbf{x}_I)$, where $\mathbf{x}_R$ and $\mathbf{x}_I$ are two independent samples from the source. The coefficient $\sqrt{p/2}$ ensures the vector $\mathbf{s}$ has power p. In this way, the transmit power constraints are satisfied. Finally, the central machine estimates the conditional dependencies of the vectors $\mathbf{x}$ using the received vectors $\mathbf{y}$.

A. Approximating the Sample Covariance

We define $\widetilde{H} = \sqrt{\frac{p}{2}}\begin{bmatrix}H_R & -H_I\\ H_I & H_R\end{bmatrix}$, $\tilde{\mathbf{x}} = \begin{bmatrix}\mathbf{x}_R\\ \mathbf{x}_I\end{bmatrix}$, $\tilde{\mathbf{y}} = \begin{bmatrix}\mathbf{y}_R\\ \mathbf{y}_I\end{bmatrix}$, and $\tilde{\mathbf{z}} = \begin{bmatrix}\mathbf{z}_R\\ \mathbf{z}_I\end{bmatrix}$. When transmitting samples through the channel, if we put two independent samples as real and imaginary parts of the vector $\tilde{\mathbf{x}}$, then the covariance matrix of $\tilde{\mathbf{x}}$ becomes

$$Q_{\tilde{x}} = \begin{bmatrix}Q_x & 0\\ 0 & Q_x\end{bmatrix}. \tag{33}$$

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on February 07,2026 at 01:36:22 UTC from IEEE Xplore. Restrictions apply.

On the other hand, according to equation (32), we have

$$Q_{\tilde{x}} = \widetilde{H}^{-1} Q_{\tilde{y}} \widetilde{H}^{-T} - \sigma_z^2 \widetilde{H}^{-1} \widetilde{H}^{-T}, \tag{34}$$

where $Q_{\tilde{x}}$ and $Q_{\tilde{y}}$ is the covariance matrix of $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{y}}$, respectively. By substituting the sample covariance matrix of $\tilde{\mathbf{y}}$ into the above expression, we obtain an approximation for the sample covariance matrix of $\tilde{\mathbf{x}}$, as

$$S_{\tilde{x}} = \widetilde{H}^{-1} S_{\tilde{y}} \widetilde{H}^{-T} - \sigma_z^2 \widetilde{H}^{-1} \widetilde{H}^{-T}. \tag{35}$$

In following lemma, we obtain an upper bound on the error probability of the $S_{\tilde{x}}$ in estimating $Q_{\tilde{x}}$.

Lemma 3: Given n i.i.d. samples of the vector $\tilde{\mathbf{y}}$. Then, for the sample covariance matrix $S_{\tilde{x}}$ in (35) *we have*

$$\Pr\left(|(S_{\tilde{x}} - Q_{\tilde{x}})_{jk}| \geq \delta\right) \leq 4 \exp\left(\frac{-n\delta^2}{c^2}\right), \tag{36}$$

for all $\delta \in \left(0, c/\sqrt{2}\right)$, where c is a constant given by

$$c = 40\sqrt{2}\left(1 + \sigma_z^2 \max_i (\widetilde{H}^{-1}\widetilde{H}^{-T})_{ii}\right). \tag{37}$$

Proof: We define the random variable $\mathbf{w}$ as

$$\mathbf{w} = \widetilde{H}^{-1}\tilde{\mathbf{y}} = \tilde{\mathbf{x}} + \widetilde{H}^{-1}\tilde{\mathbf{z}}.$$

It is clear that $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, Q_w)$, where $Q_w = Q_{\tilde{x}} + \sigma_z^2 \widetilde{H}^{-1}\widetilde{H}^{-T}$. Denoting $\mathbf{w} = [w_1, \cdots, w_{2d}]^T$, $w_j/\sqrt{(Q_w)_{jj}}$ is a standard normal variable which is sub-Gaussian with parameter 1. Thus, according to Lemma 1 in [20], we have

$$\Pr\left(|(S_w - Q_w)_{jk}| \geq \delta\right) \leq 4 \exp\left\{-\frac{n\delta^2}{3200 \max_i (Q_w)_{ii}^2}\right\},$$

for $\delta \in \left(0, 40 \max_i (Q_w)_{ii}^2\right)$, where $S_w = \widetilde{H}^{-1} S_{\tilde{y}} \widetilde{H}^{-T}$ is the sample covariance over $\mathbf{w}^{(i)} = \widetilde{H}^{-1}\tilde{\mathbf{y}}^{(i)}$ samples. On the other hand, we have

$$\begin{aligned}
&\Pr\left(|\,(S_{\tilde{x}} - Q_{\tilde{x}})_{jk}\,| \geq \delta\right) \\
&= \Pr\left(\left|\left(\widetilde{H}^{-1}\left(S_{\tilde{y}} - Q_{\tilde{y}}\right)\widetilde{H}^{-T}\right)_{jk}\right| \geq \delta\right) \\
&= \Pr\left(|\,(S_w - Q_w)_{jk}\,| \geq \delta\right) \\
&\leq 4 \exp\left\{-\frac{n\delta^2}{3200 \max_i (Q_w)_{ii}^2}\right\}.
\end{aligned} \tag{38}$$

Since the diagonal entries of Q_x are equal to 1, we have

$$\begin{aligned}
\max_i (Q_w)_{ii} &= \max_i (Q_{\tilde{x}} + \sigma_z^2 \widetilde{H}^{-1}\widetilde{H}^{-T})_{ii} \\
&= 1 + \sigma_z^2 \max_i (\widetilde{H}^{-1}\widetilde{H}^{-T})_{ii}.
\end{aligned} \tag{39}$$

By substituting (39) into (38), the error bound (36) is obtained. □

In order to obtain an estimation for Q_x, we define following estimator

$$\widehat{S}_x = \frac{1}{2}\left(\begin{bmatrix} I_d & 0_d \end{bmatrix} S_{\tilde{x}} \begin{bmatrix} I_d \\ 0_d \end{bmatrix} + \begin{bmatrix} 0_d & I_d \end{bmatrix} S_{\tilde{x}} \begin{bmatrix} 0_d \\ I_d \end{bmatrix}\right), \tag{40}$$

where 0_d is a $d \times d$ zero matrix. In the following lemma, an error bound is obtained for this estimator.

Lemma 4: For $\widehat{S}_x$ defined in (40)*, we have*

$$\Pr\left(|(\widehat{S}_x - Q_x)_{jk}| \geq \delta\right) \leq 8 \exp\left(\frac{-n\delta^2}{2c^2}\right), \tag{41}$$

where $\delta \in \left(0, c/\sqrt{2}\right)$ and c is defined by (37).

Proof: From (40), it is clear that

$$(\widehat{S}_x)_{jk} = \frac{1}{2}\left((S_{\tilde{x}})_{jk} + (S_{\tilde{x}})_{(d+j)(d+k)}\right).$$

Thus, we have

$$\begin{aligned}
&\Pr\left(|(\widehat{S}_x - Q_x)_{jk}| \geq \delta\right) \\
&= \Pr\left(|(S_{\tilde{x}})_{jk} + (S_{\tilde{x}})_{(d+j)(d+k)} - 2(Q_{\tilde{x}})_{jk}| \geq 2\delta\right) \\
&\leq \Pr\left(|(S_{\tilde{x}})_{jk} - (Q_{\tilde{x}})_{jk}| + |(S_{\tilde{x}})_{(d+j)(d+k)} - (Q_{\tilde{x}})_{jk}| \geq 2\delta\right) \\
&\leq \Pr\left(|(S_{\tilde{x}})_{jk} - (Q_{\tilde{x}})_{jk}| \geq \delta\right) \\
&\quad + \Pr\left(|(S_{\tilde{x}})_{(d+j)(d+k)} - (Q_{\tilde{x}})_{jk}| \geq \delta\right) \\
&\leq 8 \exp\left(\frac{-n\delta^2}{2c^2}\right),
\end{aligned}$$

where the last inequality is obtained by the error bound of Lemma 3 for the sample size $n/2$. □

Note that the matrix $\widehat{S}_x$ in (40) is not necessarily positive semi-definite. But this does not affect the convexity of the optimization problem (3).

By substituting $\widehat{S}_x$ from (40) into the (3), we can solve the ℓ_1 regularized maximum likelihood problem and obtain a sparse solution for the precision matrix Θ_x. Similar to Theorem 1, we can guarantee that the Uncoded method can correctly recover the underlying graph structure with high probability.

Theorem 2: Consider a normal distribution satisfying Assumption 1 with parameter $\alpha \in (0, 1]$. Let $\widehat{\Theta}_x$ be the solution of the log-determinant program (3) *with sample covariance $\widehat{S}_x$ in* (40) *and regularization parameter $\lambda_n = (8c/\alpha)\sqrt{\frac{2}{n}\ln\frac{8}{\epsilon}}$ for any $0 < \epsilon \leq d^{-2}$.*

(a) If the sample size is lower bounded as

$$n > C_{\text{uncoded}}^2 \ln\frac{8}{\epsilon}, \tag{42}$$

where

$$C_{\text{uncoded}} = 6\sqrt{2}c\ \Delta\left(1 + \frac{8}{\alpha}\right)\kappa_\Sigma^3 \kappa_\Gamma^2,$$

then with probability at least $1 - d^2\epsilon$, the edge set specified by $\widehat{\Theta}_x$ is a subset of the true edge set.

(b) If the sample size satisfies the lower bound

$$n > T_{\text{uncoded}}^2 \ln\frac{8}{\epsilon}, \tag{43}$$

where

$$T_{\text{uncoded}} = \max\left\{2\sqrt{2}c\left(1 + \frac{8}{\alpha}\right)\kappa_\Gamma \theta_{\min}^{-1},\ C_{\text{uncoded}}\right\},$$

then,

$$\Pr\left(\mathcal{M}(\widehat{\Theta}_x; \Theta_x)\right) \geq 1 - d^2\epsilon. \tag{44}$$

Proof: Similar to Theorem 1, we need to combine Lemma 4 and Lemma 2. According to Lemma 4, the function $f(n, \delta)$ defined in Lemma 2 is obtained as

$$f(n, \delta) = 8 \exp\left(\frac{-n\delta^2}{2c^2}\right),$$

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on February 07,2026 at 01:36:22 UTC from IEEE Xplore. Restrictions apply.

where $\delta \in (0, c/\sqrt{2})$. Hence, we have

$$\nu_* = \frac{\sqrt{2}}{c}, \tag{45}$$

$$\bar{\delta}(n, \epsilon) = c\sqrt{\frac{2}{n} \ln \frac{8}{\epsilon}}, \tag{46}$$

$$\bar{n}(\delta, \epsilon) = \frac{2c^2}{\delta^2} \ln \frac{8}{\epsilon}. \tag{47}$$

By substituting the above expressions into Lemma 2 and using inequalities (29) and (30), the theorem results are obtained. □

Similar to the Signs method, the above theorem shows the proposed Uncoded method can recover the underlying graph structure if the sample complexity scales with $\ln(1/\epsilon)$. Here, the constants C_{uncoded} and T_{uncoded}, in addition to the model parameters, depend on the channel parameters H and σ_z^2. However, for many channel parameters, we have $C_{\text{uncoded}} \geq C_{\text{sign}}$ and $T_{\text{uncoded}} \geq T_{\text{sign}}$, which indicates that the sample complexity required for the Uncoded method is typically larger than of the Signs method. This is supported by the experiments provided in Section V.

Remark 2: Since the channel has real and imaginary parts, 2 samples can be transmitted by each channel access (i.e. each local machine can transmit n samples by $n/2$ channel uses). Therefore, if the sample generation rate at the source is less than or equal to twice of the channel's rate, the machines can transmit all samples without any sample loss.

Although we have assumed the number of receivers at the central machine is equal to the dimension of the normal distribution, the proposed method can be extended to the high-dimensional case using *time sharing* of the channel. In this scenario, the central machine at each step requests some portion of the local machines to transmit their data. After receiving data from all local nodes, the central machine can adopt the proposed method to recover the underlying graph structure.

V. Experiments

In this section, the performance of our proposed methods is evaluated by performing several experiments. In our simulations,[2] the *glasso* package [16] is used to solve ℓ_1-regularized MLE of the precision matrix. This package is based on the block coordinate descent algorithm proposed by [38].

Denoting the bit rate of the channel at local machine j by R_j, the achievable bit rates for all machines are characterized by [39]

$$\sum_{j \in V} R_j \leq \lg \det \left(\frac{p}{\sigma_z^2} H_V^H H_V + I_{|V|} \right), \quad \forall\, V \subseteq \{1, \cdots, d\}, \tag{48}$$

where H is the channel gains matrix, H_V is the sub-matrix of H that includes the rows and columns indexed by V, H^H is the Hermitian of H, $I_{|V|}$ is the $|V| \times |V|$ identity matrix, and $\lg(\cdot)$ is the logarithm function in base 2.

In order to have a fair comparison between Signs and Uncoded methods, we have used identical parameters for the channel. More precisely, we assume $H = I$ which ensures all the local machines have identical bit rates. We set the bit rate of each local machine, i.e. R_j, to 2 bits. Thus, in the proposed methods, each local machine can transmit n bits by $n/2$ channel uses. According to (48), in order to achieve the bit-rate of 2, the signal to noise ratio (SNR) should set to 3 (i.e. $\frac{p}{\sigma_z^2} = 3$).

In the first experiment, we have employed the True Positive and False Positive Rates (TPR and FPR, respectively) as our performance measures. TPR is defined as the percentage of the predicted edges (non-zero off-diagonal entries in the precision matrix) that correctly detected. Similarly, FPR is the percentage of the predicted non-edges (zero entries in the precision matrix) that incorrectly detected. Sparse random precision matrices are generated as follows. First, we generate a random sparse graph with a fixed probability of the edge presence, say 0.1, and also set its maximum node degree to $\Delta = 5$. Then, we choose edge weights uniformly in $[-1, 1]$ for the symmetric precision matrix Θ. Next, we make it a positive definite matrix by adding a scaled identity matrix, more precisely, we set $\Theta \leftarrow \Theta + |\lambda_{min}(\Theta)|I$, where $\lambda_{min}(\Theta)$ is the minimum eigenvalue of Θ and I is the identity matrix. Finally, we set $\Theta \leftarrow D\Theta D$, where D is diagonal matrix with $D_{ii} = \sqrt{\Theta_{ii}^{-1}}$. This ensures Θ^{-1} has variances equal to 1. Also, we ensure that the generated matrix satisfies Assumption 1.

Fig. 2 shows the performance of the methods as a function of the sample size n for random sparse graphs with $d = 50$ and $d = 100$. As can be seen, the Signs method outperforms Uncoded scheme. This is supported by theorems 1 and 2 where we have $C_{\text{uncoded}} \geq C_{\text{sign}}$ (or equivalently $T_{\text{uncoded}} \geq T_{\text{sign}}$). In other words, according to the theorems, for a fixed error ϵ and the same Gaussian model and channel parameters, Uncoded method requires more samples than the Signs method. Note that original data in the plots refer to recovering the structure in a situation in which the central machine has access to samples directly.

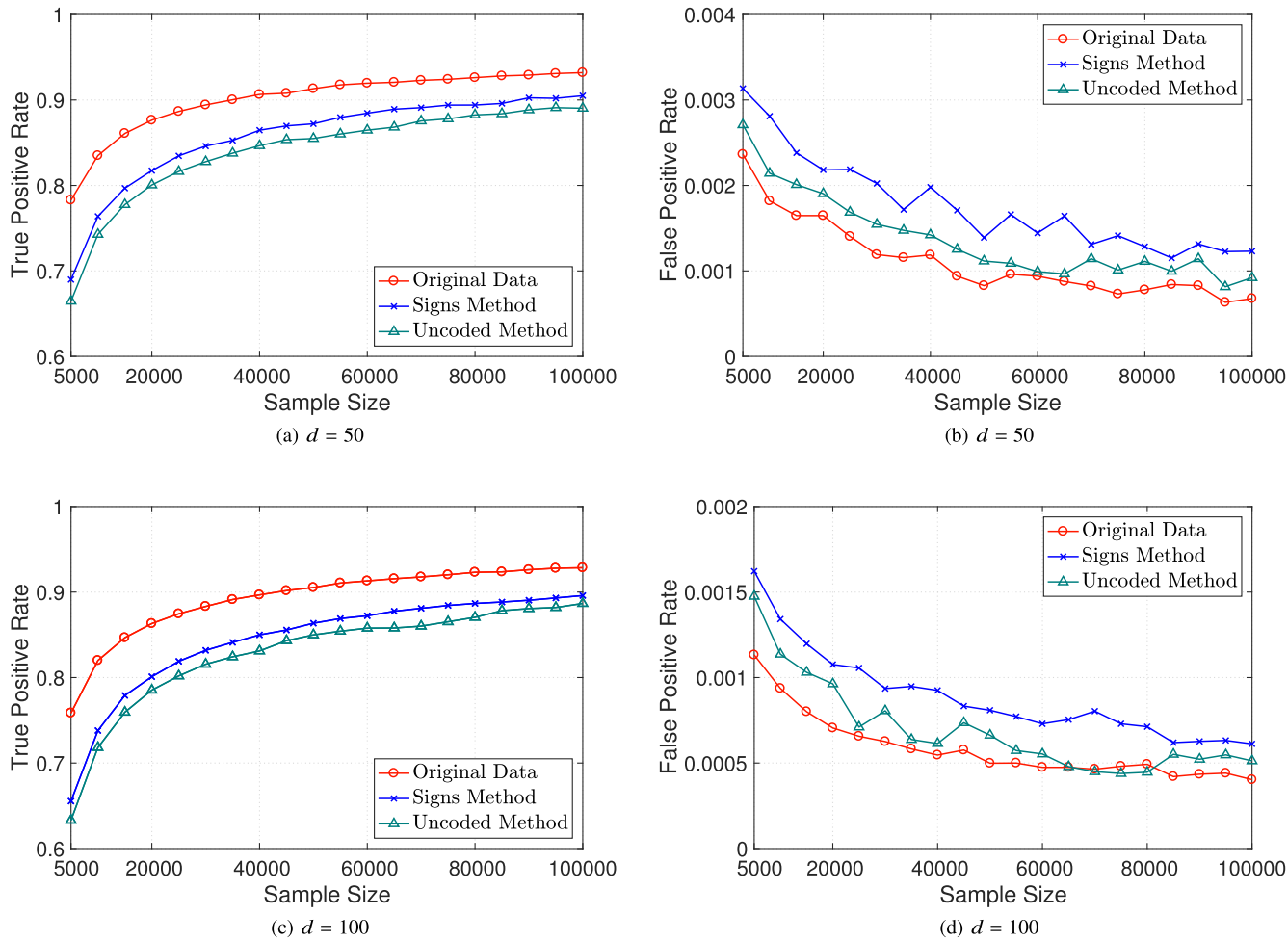
In order to demonstrate the results from theorems 1 and 2, we have plotted the probability of success, the probability of perfect recovery of edges and their signs, against sample size for Signs and Uncoded methods as well as the original data. In all of these experiments, we have set λ_n proportional to $\sqrt{\ln(d)/n}$, as provided by the theorems. For computing the success probability, we have run each method on each graph 100 times and count the number of perfect recoveries. In each experiment, we varied a specific parameter while holding all other parameters fixed to confirm the results of the theorems.

As suggested by theorems 1 and 2, the sample complexity scales with $\ln(1/\epsilon)$, where $\epsilon \leq d^{-2}$ is the probability of error. Setting ϵ to d^{-2}, leads to scaling of the sample complexity with $\ln(d)$. In the experiment of Fig. 3 and Fig. 4, we plot the probability of perfect graph recovering against the sample size n for different dimensionalities d. To show that the sample size is scaled by $\ln(d)$, we have also plotted the success curves for sample size normalized by $\ln(d)$.

In Fig. 3, we examined pruned star-shaped graphs with $\Delta = 50$, $Q_{ij} = 0.25$ for $(i, j) \in \mathcal{E}$ and $d \in \{64, 128, 256\}$. We define a pruned star-shaped graph as a graph where one

[2]The source code is available at https://github.com/ArminKaramzade/distributed-sparse-GGM

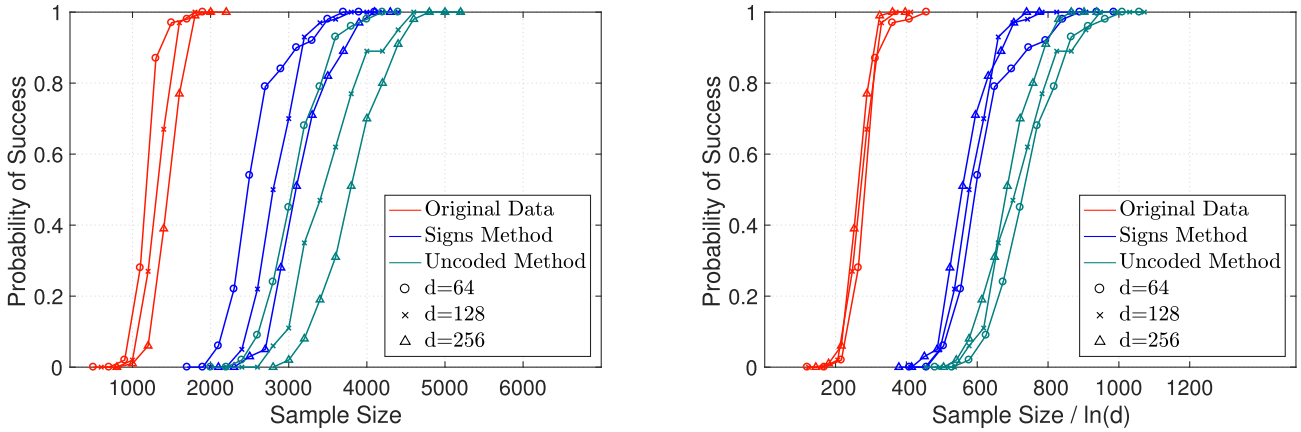
Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on February 07,2026 at 01:36:22 UTC from IEEE Xplore. Restrictions apply.



(a) d = 50 (b) d = 50

(c) d = 100 (d) d = 100

Fig. 2. TPR and FPR as a function of the sample size n for random graphs with 50 and 100 nodes. Results are average performance of methods over 20 sparse random graphs where each of them averaged 10 times with different samples.



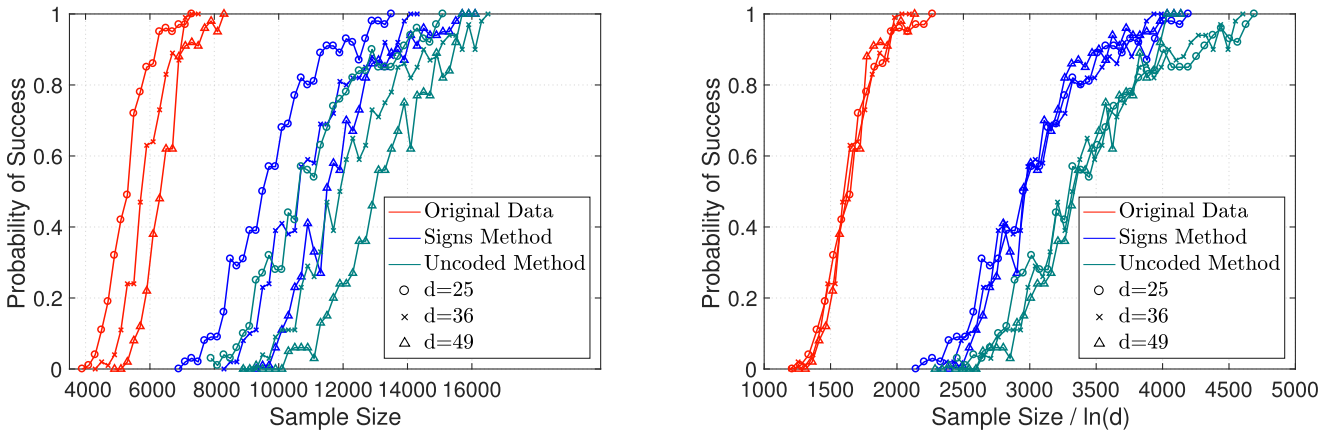
(a) Unnormalized sample size (b) Sample size normalized by $\ln(d)$

Fig. 3. The probability of perfect graph structure recovery for pruned star-shaped graphs with 64, 128 and 256 nodes.

center node is connected to Δ nodes, and the remaining nodes have a zero degree. In Fig. 3b, we have plotted the curves for the sample size normalized by $\ln(d)$. As can be seen from the normalized plots, the success probability curves are matched on each other for different values of d.

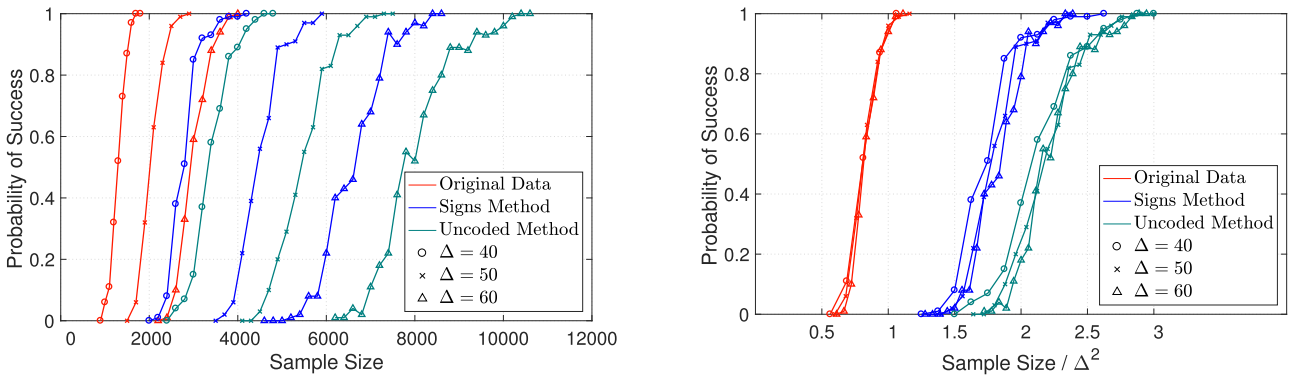
A similar experiment performed on grid-shaped graphs in Fig. 4. In this experiment, we have evaluated grids with sizes 5×5, 6×6 and 7×7. We have set all edges' weights to 0.1 in the grids. Similar to the previous experiment, the normalized probability curves (Fig. 4b) are matched over each other.

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on February 07,2026 at 01:36:22 UTC from IEEE Xplore. Restrictions apply.



(a) Unnormalized sample size (b) Sample size normalized by $\ln(d)$

Fig. 4. The probability of perfect graph structure recovery for grid-shaped graphs with sizes 5×5, 6×6 and 7×7.



(a) Unnormalized sample size (b) Sample size normalized by Δ^2

Fig. 5. The probability of perfect graph structure recovery for pruned star-shaped graphs with different maximum degrees Δ.

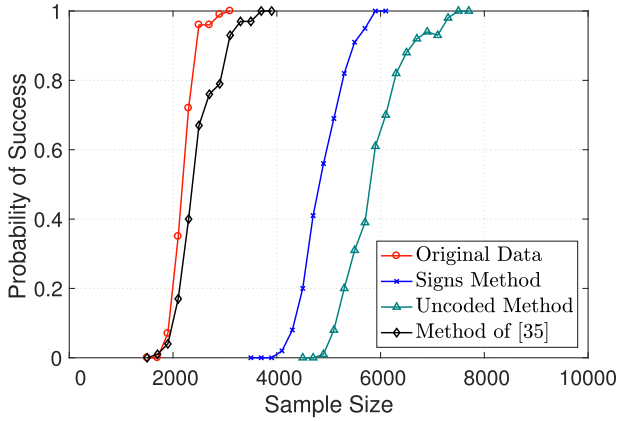


Fig. 6. Comparison of probabilities of perfect graph structure recovery among proposed methods, Original data and method of [35] for a star-shaped graph with $d = 128$.

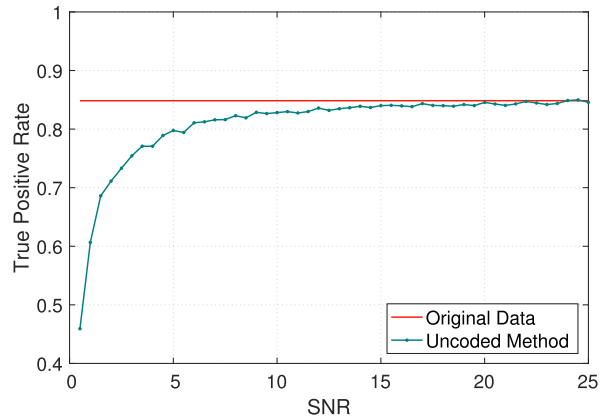


Fig. 7. TPR as a function of SNR in Uncoded method for a random graph with $d = 40$ and maximum degree $\Delta = 5$.

In the experiment of Fig. 5 we have evaluated the effect of maximum node degree Δ on the sample complexity. As suggested by theorems 1 and 2, the sample complexity scales with Δ^2 for a fixed error probability. In Fig. 5a, we have plotted the probability of success for several pruned star-shaped graphs with $d = 128$, $Q_{ij} = 0.25$ for $(i, j) \in \mathcal{E}$ and $\Delta \in \{40, 50, 60\}$. The probability curves for normalized sample size by Δ^2 are plotted in Fig. 5b which are stacked up for different values of Δ.

We have compared our methods with the proposed method by Tavassolipour *et al.* [35]. They proposed a distributed method for estimating tree-structure of GGMs. Their method has 1-bit communication cost and is only applicable for

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on February 07,2026 at 01:36:22 UTC from IEEE Xplore. Restrictions apply.

tree structured GGMs. In Fig. 6, the probability of perfect structure recovery for a star-shaped graph with $d = 128$ and $Q_{ij} = 0.2$ has been plotted. The method of [35] outperforms our proposed methods since it knows the underlying graph structure is a tree in advance.

In Fig. 7, we measure the TPR of Uncoded method for different values of the SNR. The experiment is performed on a random graph created as explained in the first experiment with $d = 40$ nodes, maximum degree of $\Delta = 5$ and $n = 10,000$. In this experiment, we have randomly generated the channel matrix H with entries drawn from i.i.d. standard normal samples. The TPR curve is averaged over 100 different channel matrices. As can be seen from Fig. 7, for SNR greater than 5, the performance of Uncoded method is very close to that of the original data. The FPR curve is not plotted, since the error values were negligible even for small SNR.

VI. Conclusion

In this paper, we have studied the sparse structure learning of GGMs where the data are distributed across multiple local machines. Two methods are proposed to send information from the local machines to the central machine, namely, Signs and Uncoded methods. We have analytically and experimentally shown that the central machine can recover the underlying graph if large enough sample sizes are transmitted to the central machine.

Our experiments show that, under the same conditions, the Signs method outperforms the Uncoded scheme. Both methods have small FPR which is close to the FPR obtained by the original data.

Acknowledgment

The authors would like to thank A. Najafi and A.-H. Saberi for their valuable comments that improved the manuscript.

References

[1] L. E. Chai, S. K. Loh, S. T. Low, M. S. Mohamad, S. Deris, and Z. Zakaria, "A review on the computational approaches for gene regulatory network construction," *Comput. Biol. Med.*, vol. 48, pp. 55–65, May 2014.

[2] M. Hecker, S. Lambeck, S. Toepfer, E. van Someren, and R. Guthke, "Gene regulatory network inference: Data integration in dynamic models—A review," *Biosystems*, vol. 96, no. 1, pp. 86–103, 2009.

[3] R. Xiang, J. Neville, and M. Rogati, "Modeling relationship strength in online social networks," in *Proc. 19th Int. Conf. World Wide Web*, 2010, pp. 981–990.

[4] Y. Li, J. Liu, J. Huang, Z. Li, and P. Liang, "Learning brain connectivity sub-networks by group- constrained sparse inverse covariance estimation for Alzheimer's disease classification," *Frontiers Neuroinform.*, vol. 12, p. 58, Sep. 2018.

[5] S. Yang, Q. Sun, S. Ji, P. Wonka, I. Davidson, and J. Ye, "Structural graphical lasso for learning mouse brain connectivity," in *Proc. 21st ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2015, pp. 1385–1394.

[6] J. R. Hershey, S. J. Rennie, P. A. Olsen, and T. T. Kristjansson, "Superhuman multi-talker speech recognition: A graphical modeling approach," *Comput. Speech Lang.*, vol. 24, no. 1, pp. 45–66, 2010.

[7] M. Bollhöfer, A. Eftekhari, S. Scheidegger, and O. Schenk, "Large-scale sparse inverse covariance matrix estimation," *SIAM J. Sci. Comput.*, vol. 41, no. 1, pp. A380–A401, 2019.

[8] R. Y. Zhang, S. Fattahi, and S. Sojoudi, "Large-scale sparse inverse covariance estimation via thresholding and max-det matrix completion," Feb. 2018, *arXiv:1802.04911*. [Online]. Available: https://arxiv.org/abs/1802.04911

[9] J. Fan, H. Liu, and W. Wang, "Large covariance estimation through elliptical factor models," *Ann. Statist.*, vol. 46, no. 4, pp. 1383–1414, 2018.

[10] R. Fan, B. Jang, Y. Sun, and S. Zhou, "Precision matrix estimation with noisy and missing data," Apr. 2019, *arXiv:1904.03548*. [Online]. Available: https://arxiv.org/abs/1904.03548

[11] A. Katiyar, J. Hoffmann, and C. Caramanis, "Robust estimation of tree structured Gaussian graphical model," Jan. 2019, *arXiv:1901.08770*. [Online]. Available: https://arxiv.org/abs/1901.08770

[12] H. Chun, M. H. Lee, S.-H. Kim, and J. Oh, "Robust precision matrix estimation via weighted median regression with regularization," *Can. J. Statist.*, vol. 46, no. 2, pp. 265–278, Jun. 2018.

[13] R. Tibshirani, "Regression shrinkage and selection via the lasso: A retrospective," *J. Roy. Stat. Soc. B, Stat. Methodol.*, vol. 73, no. 3, pp. 273–282, 2011.

[14] T. Hastie, R. Tibshirani, and M. Wainwright, *Statistical Learning With Sparsity: The Lasso and Generalizations*. Boca Raton, FL, USA: CRC Press, 2015.

[15] R. Tibshirani, "Regression shrinkage and selection via the lasso," *J. Roy. Stat. Soc. B, Methodol.*, vol. 58, no. 1, pp. 267–288, 1996.

[16] J. Friedman, T. Hastie, and R. Tibshirani, "Sparse inverse covariance estimation with the graphical lasso," *Biostatistics*, vol. 9, no. 3, pp. 432–441, 2008.

[17] N. Meinshausen and P. Bühlmann, "High-dimensional graphs and variable selection with the Lasso," *Ann. Statist.*, vol. 34, no. 3, pp. 1436–1462, 2006.

[18] C.-J. Hsieh, M. A. Sustik, I. S. Dhillon, and P. Ravikumar, "QUIC: Quadratic approximation for sparse inverse covariance estimation," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 2911–2947, 2014.

[19] J. Janková and S. van de Geer, "Confidence intervals for high-dimensional inverse covariance estimation," *Electron. J. Statist.*, vol. 9, no. 1, pp. 1205–1229, 2015.

[20] P. Ravikumar, M. J. Wainwright, G. Raskutti, and B. Yu, "High-dimensional covariance estimation by minimizing ℓ_1-penalized log-determinant divergence," *Electron. J. Statist.*, vol. 5, pp. 935–980, Jan. 2010.

[21] S. Sojoudi, "Equivalence of graphical lasso and thresholding forsparse graphs," *J. Mach. Learn. Res.*, vol. 17, no. 1, pp. 3943–3963, 2016.

[22] A. Anandkumar, V. Y. F. Tan, F. Huang, and A. S. Willsky, "High-dimensional Gaussian graphical model selection: Walk summability and local separation criterion," *J. Mach. Learn. Res.*, vol. 13, pp. 2293–2337, Aug. 2012.

[23] C. Chow and C. Liu, "Approximating discrete probability distributions with dependence trees," *IEEE Trans. Inf. Theory*, vol. IT-14, no. 3, pp. 462–467, May 1968.

[24] V. Y. F. Tan, A. Anandkumar, and A. S. Willsky, "Learning high-dimensional Markov forest distributions: Analysis of error rates," *J. Mach. Learn. Res.*, vol. 12, pp. 1617–1653, May 2011.

[25] V. Y. F. Tan, A. Anandkumar, L. Tong, and A. S. Willsky, "A large-deviation analysis of the maximum-likelihood learning of Markov tree structures," *IEEE Trans. Inf. Theory*, vol. 57, no. 3, pp. 1714–1735, Mar. 2011.

[26] Y. Zhang, J. Duchi, M. I. Jordan, and M. J. Wainwright, "Information-theoretic lower bounds for distributed statistical estimation with communication constraints," in *Proc. Adv. Neural Inf. Process. Syst.*, 2013, pp. 2328–2336.

[27] A. Xu and M. Raginsky, "Information-theoretic lower bounds on Bayes risk in decentralized estimation," *IEEE Trans. Inf. Theory*, vol. 63, no. 3, pp. 1580–1600, Mar. 2017.

[28] M. Tavassolipour, S. A. Motahari, and M. T. M. Shalmani, "Learning of Gaussian processes in distributed and communication limited systems," *IEEE Trans. Pattern Anal. Mach. Intell.*, to be published.

[29] M. El Gamal and L. Lai, "On rate requirements for achieving the centralized performance in distributed estimation," *IEEE Trans. Signal Process.*, vol. 65, no. 8, pp. 2020–2032, Apr. 2017.

[30] Y. Chen, R. S. Blum, B. M. Sadler, and J. Zhang, "Testing the structure of a Gaussian graphical model with reduced transmissions in a distributed setting," May 2019, *arXiv:1905.01376*. [Online]. Available: https://arxiv.org/abs/1905.01376

[31] A. Eftekhari, M. Bollhöfer, and O. Schenk, "Distributed memory sparse inverse covariance matrix estimation on high-performance computing architectures," in *Proc. Int. Conf. High Perform. Comput., Netw., Storage Anal.*, Nov. 2018, pp. 253–264.

[32] Z. Meng, D. Wei, A. Wiesel, and A. Hero, III, "Distributed learning of Gaussian graphical models via marginal likelihoods," in *Proc. 16th Int. Conf. Artif. Intell. Statist.*, 2013, pp. 39–47.

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on February 07,2026 at 01:36:22 UTC from IEEE Xplore. Restrictions apply.

[33] A. Wiesel and A. O. Hero, "Distributed covariance estimation in Gaussian graphical models," *IEEE Trans. Signal Process.*, vol. 60, no. 1, pp. 211–220, Jan. 2012.
[34] J. Arroyo and E. Hou, "Efficient distributed estimation of inverse covariance matrices," in *Proc. IEEE Stat. Signal Process. Workshop (SSP)*, Jun. 2016, pp. 1–5.
[35] M. Tavassolipour, S. A. Motahari, and M.-T. M. Shalmani, "Learning of tree-structured Gaussian graphical models on distributed data under communication constraints," *IEEE Trans. Signal Process.*, vol. 67, no. 1, pp. 17–28, Jan. 2019.
[36] N. Meinshausen, "A note on the Lasso for Gaussian graphical model selection," *Statist. Probab. Lett.*, vol. 78, no. 7, pp. 880–884, May 2008.
[37] R. H. Bacon, "Approximations to multivariate normal orthant probabilities," *Ann. Math. Statist.*, vol. 34, no. 1, pp. 191–198, 1963.
[38] O. Banerjee, L. El Ghaoui, and A. d'Aspremont, "Model selection through sparse maximum likelihood estimation for multivariate Gaussian or binary data," *J. Mach. Learn. Res.*, vol. 9, pp. 485–516, Mar. 2008.
[39] D. Tse and P. Viswanath, *Fundamentals of Wireless Communication*. Cambridge, U.K.: Cambridge Univ. Press, 2005.

Mostafa Tavassolipour received the B.Sc. degree from Shahed University, Tehran, Iran, in 2009, and the M.Sc. degree from the Computer Engineering Department, Sharif University of Technology (SUT), Tehran, in 2011, where he is currently pursuing the Ph.D. degree in artificial intelligence program. His research interests include machine learning, image processing, information theory, content-based video analysis, and bioinformatics.

Armin Karamzade received the B.Sc. degree in computer engineering from the Iran University of Science and Technology, Tehran, Iran, in 2017. He is currently pursuing the M.Sc. degree in artificial intelligence with the Sharif University of Technology, Tehran. His research interests include mostly machine learning, optimization, and statistic.

Reza Mirzaeifard received the B.Sc. degree in computer engineering from Shahid Beheshti University, Tehran, Iran, in 2017. He is currently pursuing the M.Sc. degree in artificial intelligence program with the Computer Engineering Department, Sharif University of Technology. His research interests include machine learning, information theory, tensor decomposition, and bioinformatics.

Seyed Abolfazl Motahari received the B.Sc. degree from the Iran University of Science and Technology (IUST), Tehran, in 1999, the M.Sc. degree from Sharif University of Technology, Tehran, in 2001, and the Ph.D. degree from the University of Waterloo, Waterloo, Canada, in 2009, all in electrical engineering. He is currently an Assistant Professor with the Computer Engineering Department, Sharif University of Technology (SUT). From October 2009 to September 2010, he was a Post-Doctoral Fellow with the University of Waterloo, Waterloo. From September 2010 to July 2013, he was a Post-Doctoral Fellow with the Department of Electrical Engineering and Computer Sciences, University of California at Berkeley. His research interests include multiuser information theory and bioinformatics. He received several awards including Natural Science and Engineering Research Council of Canada (NSERC) Post-Doctoral Fellowship.

Mohammad-Taghi Manzuri Shalmani received the B.Sc. and M.Sc. degrees in electrical engineering from the Sharif University of Technology (SUT), Tehran, Iran, in 1984 and 1988, respectively, and the Ph.D. degree in electrical and computer engineering from the Vienna University of Technology, Austria, in 1995. He is currently an Associate Professor in the Computer Engineering Department, Sharif University of Technology. His main research interests include digital signal processing, stochastic modeling, and multiresolution signal processing.

Authorized licensed use limited to: Norges Teknisk-Naturvitenskapelige Universitet. Downloaded on February 07,2026 at 01:36:22 UTC from IEEE Xplore. Restrictions apply.